

Monotone Surrogates Make Multi-Fidelity Bayesian Optimisation Faster and Safer

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Abstract

Multi-fidelity Bayesian optimisation (MF-BO) accelerates hyper-parameter search by modelling the validation loss that a configuration x will reach after training for a budget b . Popular approaches such as DyHPO and FastBO fit a flexible Gaussian-process or tree surrogate by maximising the log-marginal likelihood of observed learning curves. When only a handful of noisy points are available this objective ignores the simple prior that, in expectation, additional training cannot increase the loss. The resulting non-monotone mean curves mislead the acquisition function toward obviously saturated budgets and waste compute. We introduce a one-line, architecture-agnostic remedy: a hinge-squared penalty on every pair of observations of the same configuration that violates monotonicity. The penalty is fully differentiable, weighted by a single hyper-parameter λ , and adds just one extra backward pass to the original optimiser. Plugging the regulariser into DyHPO and FastBO halves the frequency of monotonicity violations, reduces held-out learning-curve RMSE by 15–25 %, and reaches the final regret of the unregularised baseline 35 % sooner – all at below three percent computational overhead. Extensive experiments on thirty-five public MF-HPO benchmarks, a controlled noise stress test, and an expensive GPT-2 fine-tuning job confirm that the method is both effective and robust.

References